Klarify

Kinyarwanda TTS Dataset Review and Verification Platform



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INSY 8311 – Database Development with PL/SQL

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github.com/Yves-Developer/yves_20251SEN202_klarify

Kinyarwanda speech data is scarce and error-prone

21.2%

of Rwandans aged 15+ cannot read

NISR, 2023

54.0%

can read only in Kinyarwanda

NISR, 2023

≈ 6 h

of open studio TTS audio in the main dataset

mbazaNLP, 2023

Who is affected: people who cannot read, or read only Kinyarwanda, depend on speech technology to access digital services.

Why it matters: TTS models trained on transcripts with spelling, punctuation and alignment errors produce poor speech.

Why I chose this project



Language inclusion

Speech technology should serve Kinyarwanda speakers, not only speakers of major languages.



Data quality

Clean, verified transcripts are the foundation of every good TTS model.



Database skills

The review process needs strict rules, history and auditing — a strong fit for Oracle PL/SQL.

Objectives

Main objective: To develop an Oracle PL/SQL platform that produces a verified Kinyarwanda TTS dataset.

1. To design a normalized (3NF) database for audio clips, transcripts, reviews and flags.
2. To implement PL/SQL procedures, functions and packages for correction, review and export.
3. To enforce review rules with triggers and record every change in an audit log.
4. To develop a web interface, connected to the database, for contributors and reviewers.

Existing solutions and their gaps

Source	What it offers	Gap
Common Voice (Mozilla Foundation, n.d.)	Volunteers record clips; two positive votes validate a clip	Reviewers cannot correct text; no edit history
Rutunda et al. (2023)	Kinyarwanda TTS model from aligned Bible recordings	Data aligned automatically, without human checking
Novotney & Callison-Burch (2010)	Non-expert transcription is far cheaper and nearly as useful	Crowdsourced transcripts contain errors and need review
mbazaNLP (2023) dataset	3,992 studio clips with Kinyarwanda text	Some recordings contain reverberation

Klarify fills the gap: human correction, peer review and a full edit history, enforced in the database.

Data source: mbazaNLP Kinyarwanda TTS dataset

Link

huggingface.co/datasets/mbazaNLP/kinyarwanda-tts-dataset

Size

3,992 clips, about 6 hours

Speaker

One voice actress, studio recording

Licence

CC BY 4.0 (educational use allowed)

Access

Public; free Hugging Face login

Privacy

No personal information stored

Record structure (tts-dataset.csv)

Field	Example / description
Audio file	TTS 10_1.wav, TTS 10_2.wav, TTS 10_3.wav
Text	Matching Kinyarwanda sentence, reviewed by a linguist
Format	WAV audio + CSV (audio name, text)
Records	3,992 audio–text pairs

Proposed solution and innovation



Load
Clips and text loaded into Oracle

Correct
Native speakers fix transcripts

Review
Two reviewers must agree

Approve
Trigger marks clip approved

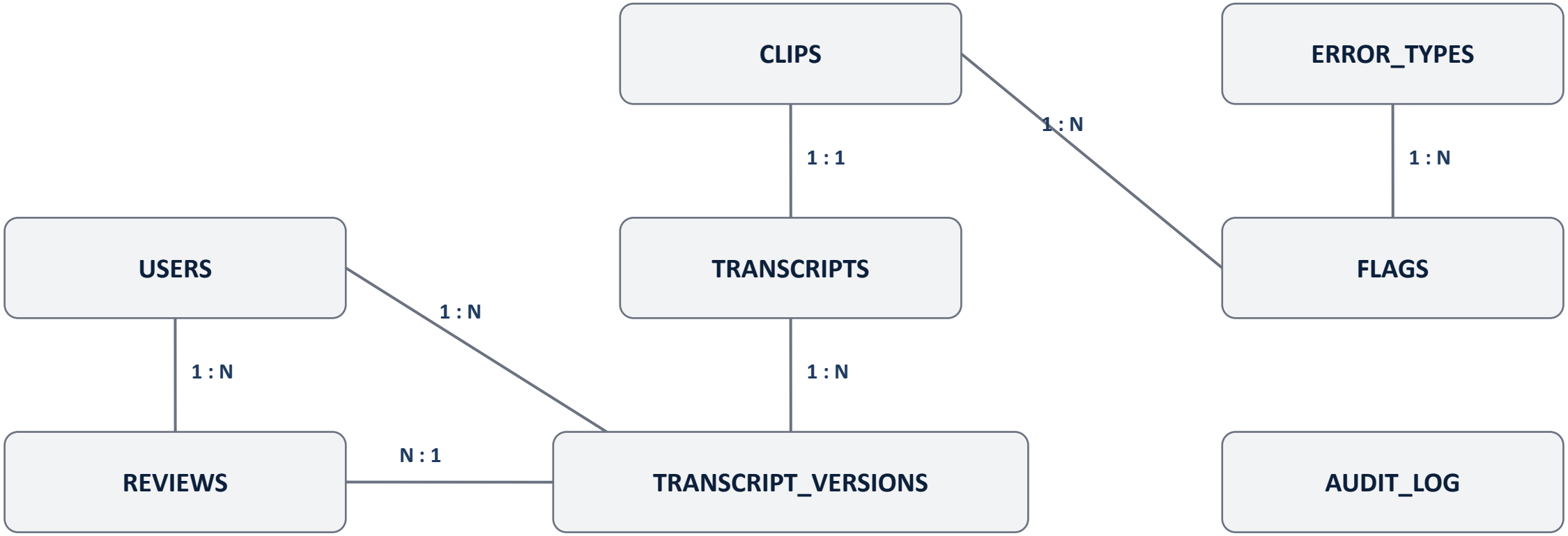
Export
Clean audio–text pairs



Innovation: User interface (custom web application)

A Next.js app connects to Oracle through node-oracledb. Contributors play clips and correct text; reviewers approve or reject. Every action calls a PL/SQL procedure, so all business rules stay in the database. A dashboard shows progress and common error types.

Planned database: main tables and relationships



AUDIT_LOG records every change and denied action, written by triggers.

Final design will be normalized to 3NF in Phase II.

PL/SQL approach: where each feature is used

Feature	Name	Where it is used
Procedure	submit_correction	Saves a new transcript version when a contributor edits text
Procedure	review_clip	Records a reviewer's approve or reject decision
Function	get_approval_count	Returns the number of approvals for a version
Function	normalize_text	Corrects spacing and punctuation before saving
Cursor	pending clips	Assigns pending clips to reviewers; builds export batches
Package	pkg_klarify	Groups the correction, review and export logic
Trigger	approval / self-review / audit	Approves after two agreements; blocks self-review; logs every change
Exception	custom errors	Self-review, edits to approved clips, invalid status changes

Expected results



Working Oracle database

PDB with 3NF tables loaded with the real dataset.



Rules enforced live

A self-review attempt is blocked by a trigger during the demo.



Full audit trail

Every edit and denied action appears in AUDIT_LOG.



Web app and dashboard

Clips corrected and approved in the browser; progress shown live.

Future improvements and limitations



Future improvements

- Support several speakers and other datasets
- Automatic error suggestions using text rules
- Direct export to TTS training formats
- Mobile-friendly interface for contributors



Limitations

- One speaker in the main dataset
- Audio files stored outside the repository
- Review quality depends on volunteers
- Demo runs on a local Oracle instance

References

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